



# Shriyansh Raj

Roll No.: 230103094

B.Tech - Mechanical Engineering

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Github

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## EDUCATION

| Degree/Certificate | Institute/Board                          | CGPA/Percentage | Year           |
|--------------------|------------------------------------------|-----------------|----------------|
| B.Tech             | Indian Institute of Technology, Guwahati | 8.06 (Current)  | 2023 - Present |
| Senior Secondary   | Central Board of Secondary Education     | 93.4%           | 2023           |
| Secondary          | Central Board of Secondary Education     | 94.2%           | 2021           |

## EXPERIENCE

- **Data Science Intern | AGC Asia Pacific** Dec. 2025 - Feb. 2026
  - Architected a multi-phase **NLP**-based research paper **recommendation system** on **104,624** papers, and **10** researcher needs
  - Diagnosed Positive-Unlabelled failure in **Learning-to-Rank**; redesigned labels to explicit-feedback-only, removing **86%** noise
  - Trained **XGBoost** binary classifier across **546** feedback labels using temporal validation, achieving **AUC 0.83-0.93** on holdouts
  - Deployed Qwen2.5-14B **LLM** for Problem-Method semantic matching; **1,600** expert-graded papers achieved **40.1%** approval
- **ML Research Intern | Bay Area Advanced Analytics** May 2026 - Aug. 2026
  - Benchmarked **5 VLMs** including MedGemma-4B & other models across **4** medical and general-domain datasets on **7** metrics
  - Designed an **LLM-as-a-Judge** pipeline for medical VQA; validated reliability via  $\kappa = 0.696$ ,  $r = 0.81$  against a **70B** judge
  - Ran **3** ablation studies and **ECE calibration**; found architectural gap ( $p > 0.05$ ), with MedGemma **23.7 pp** better-calibrated
  - Built radiology QA flow via LLM report parsing & DICOM mapping on private clinical data; validated on **5** quality checks

## PROJECTS

- **Credit Card Fraud Detection via Supervised & Anomaly-Based Learning** Jul. 2026 - Aug. 2026  
*Personal Project* GitHub
  - Built a fraud detection pipeline on **IEEE-CIS data**, engineering entity and missingness-aware features under **11.7:1 imbalance**
  - Compared Logistic Regression, XGBoost, PyTorch **autoencoder** via temporal split, achieving **0.84 AUC-ROC/0.58 PR-AUC**
  - Translated predictions into a cost model, identified threshold cutting fraud-related costs  $\sim 20\%$ , within review-queue capacity
  - **Tech Stack**: Python, Scikit-learn, PyTorch Autoencoder, PyArrow (Parquet), Matplotlib
- **Domain-Adapted Medical LLM with RAG and QLoRA Fine-Tuning** May 2025 - Jun. 2025  
*Coding Club, IITG* GitHub
  - Built a clinical Q&A chatbot by **fine-tuning** Phi-3 Mini via **QLoRA** on **2,000** NIH samples, outperforming **4** Falcon baselines
  - Engineered **FAISS RAG** on **14.5K** MedQuAD passages; lifted BERTScore **0.920**  $\rightarrow$  **0.974**; cut NLI contradiction **21.7%**  $\rightarrow$  **7.8%**
  - Devised **5-tier** clinical eval curbing verbosity bias via **ROUGE@50tok**; **4-way** ablation found **2.1%** OOD gap on PubMedQA
  - **Tech Stack**: PyTorch, HuggingFace (Transformers, PEFT), BitsAndBytes, FAISS, SentenceTransformers
- **ML-Driven Demand Forecasting & Inventory Overstock Optimisation** May 2025 - Jun. 2025  
*Consulting & Analytics Club, IITG* GitHub
  - Identified **+11.7%** forecast over-bias as root cause of **36%** overstock in **109K+** records via **SQL** diagnostics & Python t-tests
  - Built XGBoost on **150 pairs (39 features)**, achieved **12% MAPE** and  $R^2 = 0.88$ ; improved **79%** vs baseline, **17%** vs SARIMA
  - Validated forecast bias via simulation (**recall 67.5**  $\rightarrow$  **88.3%**); found OLS price inelasticity; built **2-tier** SKU split (**K-Means**)
  - **Tech Stack**: Python (XGBoost, LightGBM, SARIMA, SHAP, K-Means, statsmodels, scipy), SQL, Tableau

## TECHNICAL SKILLS & KEY COURSES TAKEN

- **Languages & Databases**: Python, MySQL, C++\*
- **ML/DL Frameworks**: Scikit-learn, PyTorch, TensorFlow\*
- **MOOC**: Summer Analytics, Time Series Analysis
- **NLP/LLM**: Hugging Face, LangChain, FAISS
- **Data Analysis**: NumPy, Pandas, Tableau, Matplotlib
- **Courses**: ML for engineers, (MA102) Linear Algebra

## POSITION OF RESPONSIBILITY

*\*Elementary proficiency*

- **Institute Joint Secretary, Saathi Counselling Cell, IIT Guwahati** May 2025 - May 2026
  - Led Saathi's **30**-member core team, **20** coordinators, **200+** mentors to drive peer mentorship, **5+** events & wellness initiatives
  - Conducted **3-day** ice-breaking session for **1,000+** freshers, overseeing planning, activities, execution, and overall coordination
  - Launched bi-weekly '**Mental Health Mondays**' series, increasing instagram page reach by **30%** while engaging **10K+** students

## ACHIEVEMENTS & EXTRACURRICULARS

- **Bronze Medal**, Solar Panel Recycling Challenge at Inter IIT Tech Meet 13.0, by INA Solar among **23** IITs 2024
- **2nd Runner-Up**, Strategy Storm - Case Competition by C&A IIT Guwahati, among **3,800+** teams globally 2025
- **Finalist - Top 18**, Indian Case Challenge by Business Club, IIT Kharagpur, among **5,400+** teams nationally 2025
- **Round 1 Qualifier**, American Express Campus Challenge, among **17,600+** participating teams nationally 2026
- **2nd Runner-Up**, Product Decode (ProdExpo) by IIM Calcutta, competed among **500+** total teams nationally 2024
- **Silver Medal**, Startup Sprint at Kriti - Inter-Hostel Technical Competition, IIT Guwahati, among **14** hostels 2025
- **Top 6 Rank**, ProdMaster - Case competition organised by IIM Kozhikode among **1,100+** teams nationally 2024
- **Bronze Medal**, Product Perspective at Kriti - Inter-Hostel Technical Competition, IITG, among **14** hostels 2024
- **Gold Medal**, Mime for hostel team at Annual Inter-Hostel Cultural Competition - Manthan, IIT Guwahati 2024
- **Silver Medal**, Hockey for hostel team at Annual Inter-Hostel Sports Competition - Spardha, IIT Guwahati 2024
- **Bronze Medal**, Hockey for hostel team at Annual Inter-Hostel Technical Competition - Kriti, IIT Guwahati 2024